

Securing Email Communications

Internet and Web Application Security 3e - Lab 9

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Report Generated: Thursday, July 30, 2026 at 5:18 PM

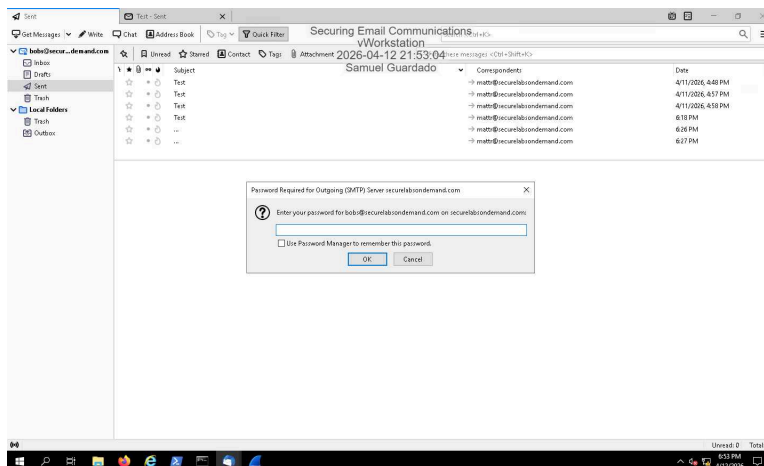
Time on Task: 1 hour, 19 minutes

Progress: 100%

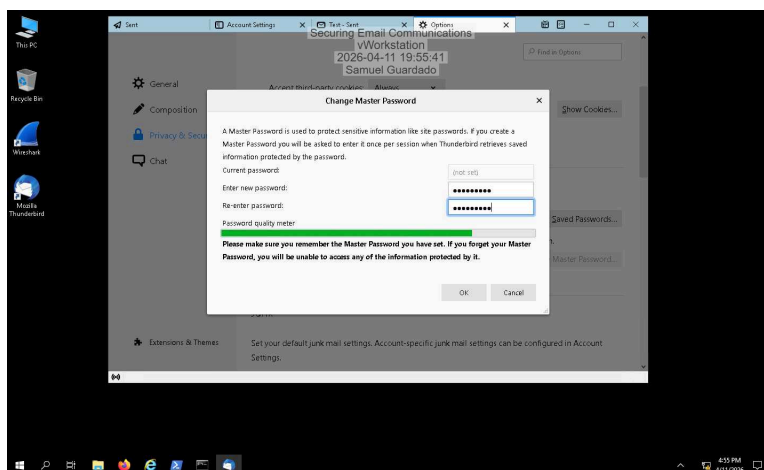
Hands-On Demonstration

Part 1: Configure Thunderbird to Secure Access on the Client

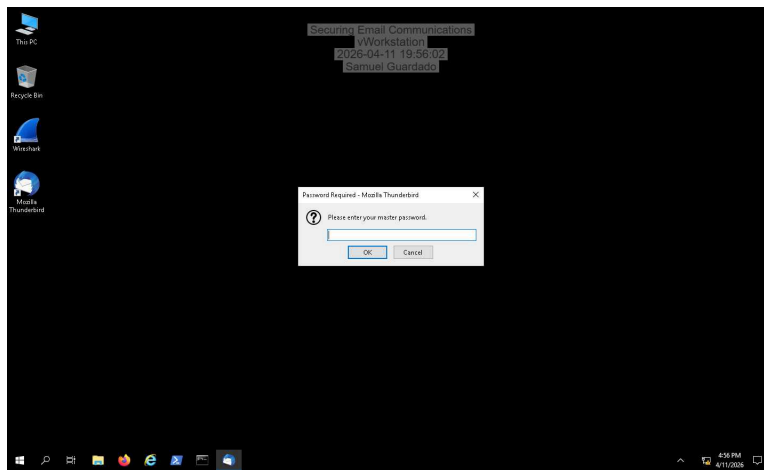
8. Make a screen capture showing the dialog box asking for the password to send an email.



17. Make a screen capture showing the new password set up including the strength meter.

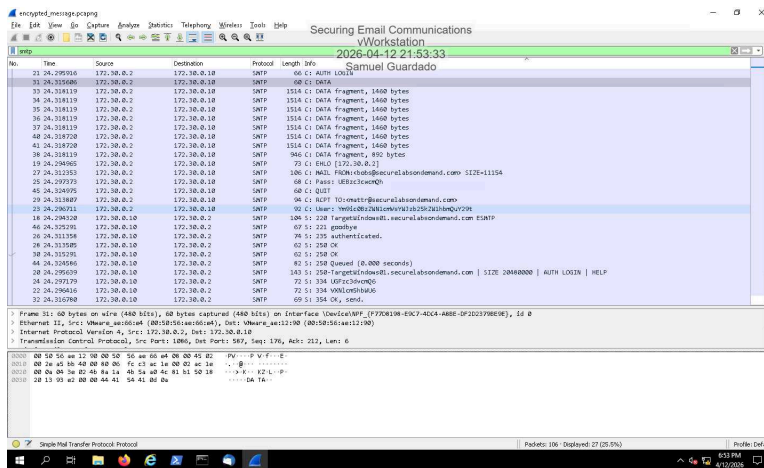


23. Make a screen capture showing the dialog asking for the master password.

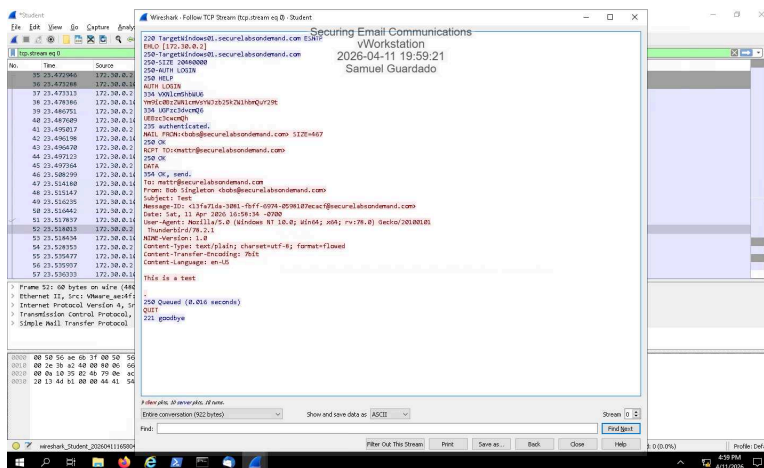


Part 2: Analyze Email Traffic with Wireshark

10. Make a screen capture showing the SMTP packets.

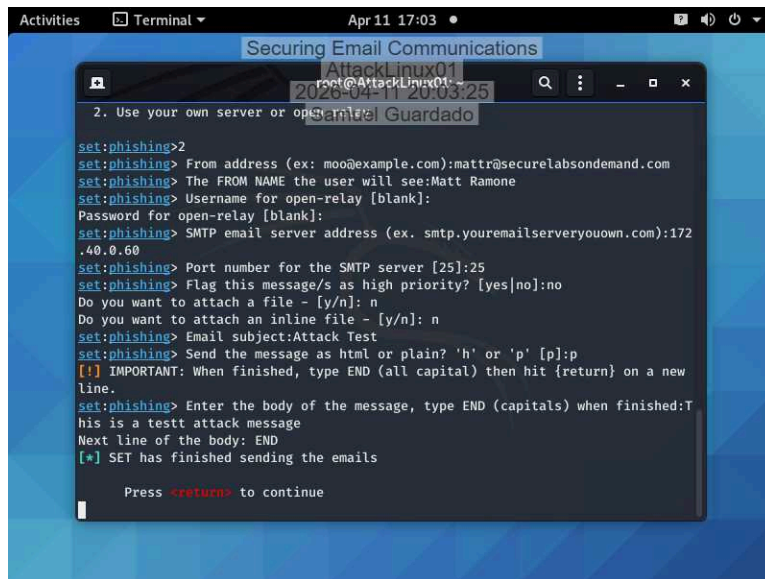


13. Make a screen capture showing the body of the email message.

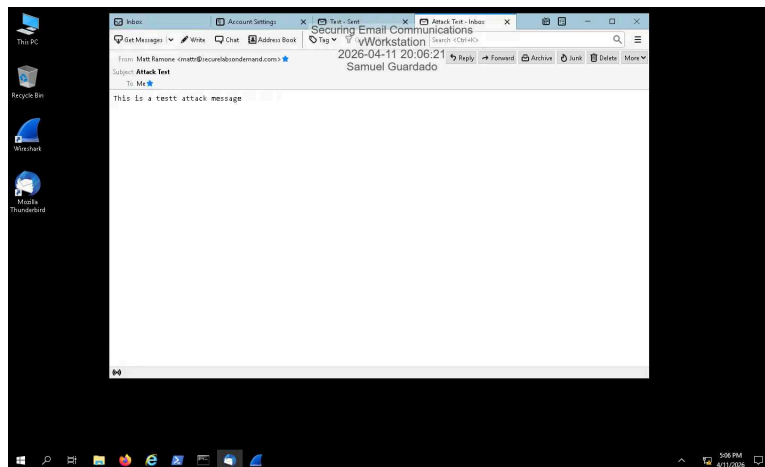


Part 3: Examine Email Headers for a Forged Message

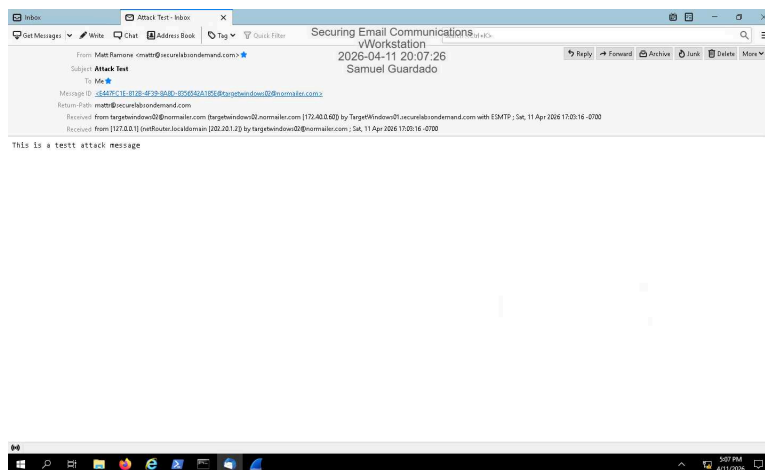
22. Make a screen capture showing the setoolkit message that the email was sent.



27. **Make a screen capture** showing the forged email including the default headers shown.



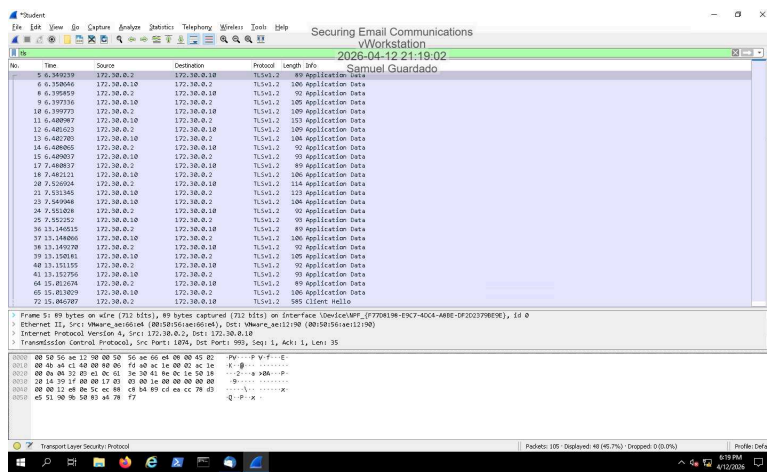
31. **Make a screen capture** showing the full headers for the forged email.



Applied Learning

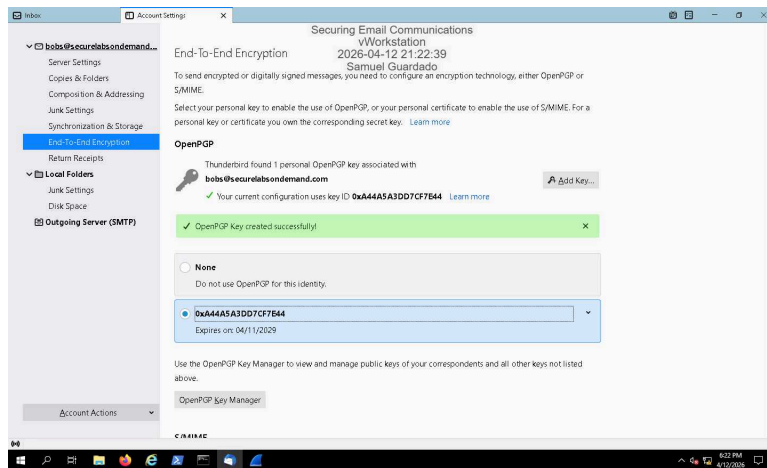
Part 1: Secure Email Communication Channels

23. **Make a screen capture** showing at least the first six packets in the packet list.

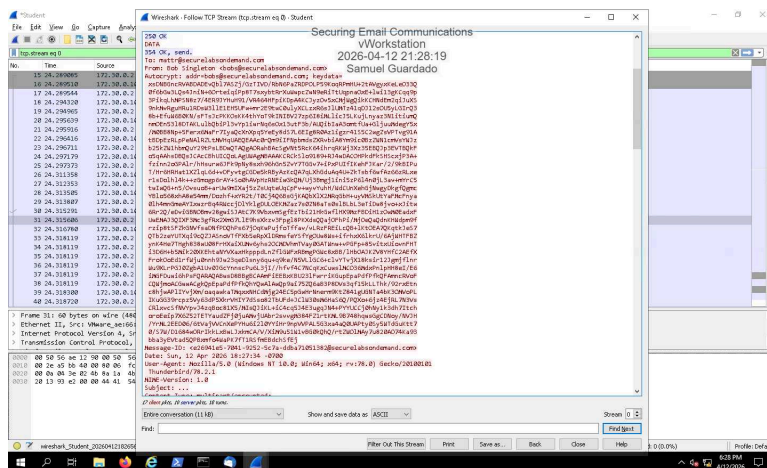


Part 2: Secure Email Communication Messages

9. Make a screen capture showing the new active OpenPGP key in the settings.



24. Make a screen capture showing the DATA fragment packets in the Wireshark packet list.



Challenge and Analysis

Part 1: Require Secure Communications

Document the changes to the Thunderbird settings needed to require encryption for all messages.

To prevent attackers from packet sniffing it is necessary to use IMAP for incoming mail that uses SSL/TLS and port 4664 or 587. Also using SMTP for outgoing mail that used SSL/TLS and port 456 or 587 to encrypt communication between the sender and receiver.

Part 2: Extract an encrypted message

Document the SMTP bytes of the encrypted message.

The SMTP bytes of the encrypted message are 44 41 54 41.